

SOHAN BISWAS

Associate Engineer - AI/ML (SDE-I)
+91 7501223507,6294142932 ◇ Bangalore, India

sohanbiswas09@gmail.com ◇ linkedin.com/in/sohan-biswas-5a9742305/ ◇ github.com/SohanBiswasSB

SUMMARY

Results-driven Software Development Engineer (SDE-I) with 2+ years of experience designing and deploying scalable backend systems on Google Cloud Platform (GCP) and Amazon Web Services (AWS). Over 1 year of focused expertise in applied AI/ML solutions, with hands-on proficiency in Django, PostgreSQL, Redis and the integration of NLP and LLM-powered features into production systems. Adept at leveraging data-driven decision-making, process automation (Python scripting, N8N), and cloud-native architectures to drive product efficiency and growth. Passionate about contributing to innovative, impact-driven projects in AI/ML engineering and backend development.

EXPERIENCE

Associate Engineer - AI/ML
Miles Education

September 9, 2024 - Present
Bangalore, India

- Led the backend development for the **Miles One** app as part of a 2-person team, implementing robust and scalable server-side logic using Django, Rest API(s), Redis and PostgreSQL.
- Designing and integrating an **LLM based chat-bot** using NLP techniques to improve student engagement with certain course transcripts navigation that aligns user interactions with company goals to enhance engagement and conversions.
- Developed and integrated multiple notification channels, including in-app and push notifications, WhatsApp alerts, and email reminders, ensuring that users stay up to date on key events and offerings.
- Built a foundational version of a command-driven **ChatBot** using **MCP** and **agent-based frameworks**, designed to generate actionable insights for internal stakeholders. I am actively enhancing this system to improve its capabilities, aiming to support data-driven decision making and align user interactions with broader business objectives.
- Handled analytics generated from app interactions, extracting insights to improve user experience, optimize app features, and refine marketing strategies.
- Independently designed and developed the registration system for **Miles Education** and its subsidiaries, for user authentication and data management to keep track of its employees around the world.
- The system ensured secure access management, centralized data handling, and seamless integration with existing enterprise applications, improving operational efficiency and security.

Digital Associate
Amazon AWS

November 8, 2021 - September 6, 2024
Bangalore, India

- **Developed automation and cost-optimization solutions using Python and AWS services**, streamlining team workflows and delivery processes, **implementing cost-effective strategies**, optimizing pricing models, and contributing to **significant revenue savings and improved operational efficiency**.
- Performed data creation and annotation for ML models including **Amazon Textract** (OCR and document parsing), **Amazon Rekognition** (image and video analysis), **Amazon Transcribe** (speech-to-text) and others.
- Utilized **Amazon SageMaker** and **SageMaker Ground Truth** to build high-quality labeled datasets, ensuring accuracy and consistency across multiple training pipelines.
- Served as **SME** for data operations across **7 AI/ML projects**, managing data sourcing, classification, Hive audits, annotation workflows, QA/MQA processes, SQL analysis, and tool testing. Led a team and ensured data integrity while maintaining detailed reports and documentation.
- Executed **SOPs** to ensure dataset quality for stakeholders, analyzed detected errors, proposed improvements, and consistently maintained **150 percent productivity benchmarks**.

SKILLS

Programming	Python, Java, C, C++, HTML5, CSS, MySQL.
Frameworks and Technologies	TensorFlow, NetCore Cloud, FCM, Weka, Redis, Postgres SQL, N8N, Tableau
Cloud Computing Platforms	AWS, GCP
Domains	AI/ML, Data Science, Forecasting, NLP, Gen AI, LLM, Time Series Analysis

EDUCATION

Master of Computer Science , Pondicherry University	2018 - 2020
Relevant Coursework: Machine Learning, Data Analysis and AI/ML Algorithms.	<i>Pondicherry, India</i>
Bachelor of Computer Science , University of Kalyani	2014 - 2017
Relevant Coursework: Java, C, C++	<i>Kalyani, India</i>

PROJECTS

Crime Prediction and Analysis	December 2019 - June 2020
Pondicherry University	<i>Pondicherry, India</i>

- Conducted comprehensive data analysis, applying advanced classification algorithms to the San Francisco Crime dataset spanning January 2003 to May 2018; produced a detailed report highlighting patterns and trends, supporting data driven strategies for targeted crime prevention initiatives.
- Predictive models and forecasting were made using AI/ML approach and libraries like TensorFlow, LSTM, Scikit-learn, Keras, Prophet, Pandas, Spicy, Numpy, etc. for 42+ classifications of various crime predictions.
- Using Matplotlib, Folium, and other visualization libraries to create data-driven insights on crime patterns, predict incidents, and identify hot spots with heatmap analysis.

Fertility Analysis	April 2019 - June 2019
Pondicherry University	<i>Pondicherry, India</i>

- Data Mining was done using Machine Learning libraries and algorithms for detection of fertility rate for four seasons and patterns.
- Predictive models were made using ML approach and tools like Scikit-learn, Seaborn, Statsmodels, etc.
- Data Analysis and representations were done in Weka and Python and a comprehensive report was generated.

Image Comparison Software	October 2016 - March 2017
University of Kalyani	<i>Kalyani, India</i>

- An application written to track the difference between two images using 8-bit RGB color codes and pointing them out, with 100 percentage accuracy rate of detecting difference.
- Built the software using Java on MATLAB and created an UI application for it for interactions.

AWARDS, CERTIFICATIONS AND SUMMER SCHOOLS

- Prompt Engineering for LLM/ChatGPT, DeepLearning.ai and Coursera.
- Summer school on Introduction to Machine Learning, IIT Kanpur.
- Time Series Analysis and Forecasting using Python, Udemy.
- Python and Statistics for Financial Analysis, Coursera.
- HTML 5 Programming, NIIT.
- 5x “Prime Player” award for Productivity, Quality, Process Compliance and Delivery expectations in AWS org.